

Multilinear Interpolation

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In these notes I try to give an overview of some different approaches to multilinear interpolation. They are framed in the setting of [BL03, Chapter 2], which provides explanation for the notation and definitions used here and in [Jan88]. This is the first time I have been in contact with these concepts, and the presence of errors is a nontrivial possibility!

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1 What we want?

From the meeting discussion (17-06-2026), it seems that we want a version of interpolation for n -sublinear functions that starts with $n + 1$ points P_j ,

$$P_j = (p_1^{(j)}, \dots, p_{n+1}^{(j)}),$$

where in each point the operator has restricted weak type (corresponding to boundedness as an operator $T: \prod_{k=1}^n L^{p_k^{(j)},1} \rightarrow L^{p_{n+1}^{(j)},\infty}$), and concludes that for any point $P = (p_1, \dots, p_{n+1})$ in the interior of the convex hull of the P_j , the function is bounded as an operator $T: \prod_{k=1}^n L^{p_k, p_k} \rightarrow L^{p_{n+1}, p_{n+1}}$. Notice that we think it's sufficient if the conclusion is restricted to Lebesgue spaces $L^p = L^{p,p}$ here. We also want it to work for cases where some of the p are in $(0, 1)$, corresponding to quasi-Banach spaces.

Taking [Gra14, Theorem 7.2.2] as the most general statement, we have a limitation

$$\frac{1}{p_{n+1}} \leq \sum_{k=1}^n \frac{1}{p_k}.$$

More generally, if $s_{n+1}^{-1} \leq \sum_{k=1}^n s_k^{-1}$ (and no restrictions on P other than it being in the interior of the hull), we can interpolate the function into a bounded operator

$$T: \prod_{k=1}^n L^{p_k, s_k} \rightarrow L^{p_{n+1}, s_{n+1}}.$$

The question is, is the statement on Grafakos the one that should be formalized? The proof is very long (20 pages) and calculational (involving matrices, many kinds of inequalities), and it is restricted to Lorentz spaces. The proof is also not entirely self-contained, and cites some exercises (pp. 497, 505, 508). (note (Lua): If AI could do it, I would be very happy, but for humans it looks like that long proof is a pain. It also does not look like the sort of theory-building proofs that we usually have in Mathlib.)

The abstract linear interpolation theory is a construction that seems well-known and has more than one reference, so in that sense it seems that there are less chances of running into problems or gaps. The argument proceeds by constructing a certain object (the K -functional) and proving properties about it, so it is more modular and easier to make modifications (see the the discussion about subadditivity below). It also provides interpolation between spaces other than Lorentz spaces (and it also includes the quasi-Banach Lorentz cases, see note 2.3).

Therefore, it's natural to question how easy it would be to get a multisublinear result as general as Grafakos' starting with the abstract linear theory, or if we could make some concessions in the statement for that goal while still keeping it useful in practice.

First, about having subadditive functions instead of linear functions in the general method, this seems

2 Introduction

We introduce the problem by considering two cases.

2.1 Interpolating between two points

This is the case considered in [Jan88]. It also seems to be used as a last step in the convex hull case, to be covered on section 2.2.

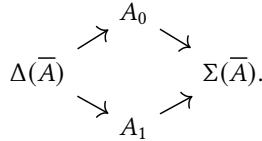
A *Banach couple* is a pair $\bar{A} = (A_0, A_1)$ of Banach spaces that are subspaces of a common Hausdorff topological vector space. In this context, it makes sense to take the intersection $\Delta(\bar{A}) = A_0 \cap A_1$, which is a Banach space with the norm

$$\|a\|_{\Delta(\bar{A})} = \max\{\|a\|_{A_0}, \|a\|_{A_1}\},$$

and also their sum $\Sigma(\bar{A}) = A_0 \cup A_1$, which is also a Banach space when supplied with the norm

$$\|a\|_{\Sigma(\bar{A})} = \inf_{a=a_0+a_1} (\|a_0\|_{A_0} + \|a_1\|_{A_1}).$$

The picture looks like this:



For the rest of the section, we let

$$\bar{A}_1 = (A_0^{(1)}, A_1^{(1)}), \dots, \bar{A}_n = (A_0^{(n)}, A_1^{(n)})$$

and $\bar{B} = (B_0, B_1)$ be Banach couples.

2.1.1 Complex method

Our reference [Jan88] writes that “it is well known that the real method of interpolation does not behave as well as the complex method for multi-linear operators”. The shortcoming of the complex method, however, is that it does not give us the Lorentz space estimates that the real method is able to give. Let’s first revise how the multi-linear result looks like for the complex method, so we can compare it with the real one.

Theorem 2.1 ([BL03, Theorem 4.4.1]). *Let*

$$T: \Delta(\bar{A}_1) \times \cdots \times \Delta(\bar{A}_n) \rightarrow \Delta(\bar{B}),$$

be a multilinear function, and assume

$$\|T(a_1, \dots, a_n)\|_{B_i} \leq K_i \|a_1\|_{A_i^{(1)}} \cdots \|a_n\|_{A_i^{(n)}}$$

for $a_j \in A_i^{(j)}$ and both $i = 0, 1$. Then for all $0 \leq \theta \leq 1$, T may be uniquely extended to a multilinear map

$$T: [\bar{A}_1]_\theta \times \cdots \times [\bar{A}_n]_\theta \rightarrow [\bar{B}]_\theta,$$

with norm at most $K_0^{1-\theta} K_1^\theta$.

This is enough for multilinear interpolation between L^p spaces:

Note 2.2 ([BL03, Section 5.1]). *If our Banach couple $\bar{A} = (A_0, A_1)$ is given by two L_p spaces $A_0 = L_{p_0}$ and $A_1 = L_{p_1}$ with $1 \leq p_0, p_1$, then for $0 \leq \theta \leq 1$ the complex interpolation space $[\bar{A}]_\theta$ is the Lebesgue space L_p where $p^{-1} = (1 - \theta)p_0^{-1} + \theta p_1^{-1}$.*

However, it does not give us Lorentz space estimates. For that, we have to turn to the next tool.

2.1.2 Real method(s)

The main reason to consider the real method, then, will be the following:

Note 2.3 ([BL03, Theorem 5.3.1]). *If our Banach couple $\bar{A} = (A_0, A_1)$ is given by two Lorentz spaces $A_0 = L_{p_0, q_0}$ and $A_1 = L_{p_1, q_1}$ where $p_0, p_1, q_0, q_1 \in (0, \infty]$, then for $0 < \theta < 1$ and $q \in (0, \infty]$, the real interpolation space $(\bar{A})_{\theta, q}$ is the Lorentz space $L_{p, q}$ where $p^{-1} = (1 - \theta)p_0^{-1} + \theta p_1^{-1}$ when $p_0 \neq p_1$, and the same is true for $p_0 = p_1$ provided that $q^{-1} = (1 - \theta)q_0^{-1} + \theta q_1^{-1}$.*

Note 2.4. $L_{p, p} = L_p$.

For the case of the real method, the first result mentioned in [Jan88] is the following:

2.2 Interpolating on the interior of a convex set

We assume that the Marcinkiewicz real interpolation theorem for linear operators is available, and we see how it can generalize to the multilinear case.

[LP64]

3 Questions for Thiele

- how/what kind of multilinear interpolation is used in the multiple ergodic averages (the new result)
- how important are the bad tuples (quasi banach estimates) in practice? do we need them formalized?
- **phrase this question better** how important is it to have the optimal bounds on the weak parameters of the multilinear interpolation? does a weaker bound suffice if it means an easier proof?

4 Applications

4.1 Usage in the commuting ergodic averages paper

References

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